

Exercise 3: SQL Case statements

Questions

1. Table: Products

Product id	Product-name	Price	Price category
1	Laptop	1200.00	Expensive
2	Phone	800	Midrange
3	Keyboard	45	Budget
4	Monitor	300	Midrange
5	Mouse	25	Budget

Expensive if price > 1000

Midrange if price between 100 and 1000

Budget if price < 100

E.C - Product name column 1
Price column 2
- Price category

```
SELECT product-name,  
price,
```

```
CASE
```

```
WHEN price < 100 THEN 'Budget'
```

```
WHEN Price between 100 and 1000 THEN 'Midrange'
```

```
WHEN price > 1000 THEN 'Expensive'
```

```
END AS price-category
```

```
FROM products;
```


2.

Order id	Customer name	Amount	Amount category
1	Alice	150	Low value
2	Bob	560	Medium value
3	Charlie	999.99	Medium value
4	Diana	45.50	Low value
5	Ethan	1200.00	High value

High value for orders ≥ 1000

Medium value for 500-999.99

Low value for orders < 500

EC - customer_name

Amount

Amount value category

```
SELECT customer_name,  
        Amount,
```

```
CASE
```

```
WHEN amount < 500 THEN 'Low value'
```

```
WHEN amount 500-999.99 THEN 'Medium value'
```

```
WHEN amount  $\geq 1000$  THEN 'High value'
```

```
END AS order_value_category
```

```
FROM orders;
```


Employees

Employee id	Emp name	Department	Salary	Position level
1	John	IT	85000	Senior IT
2	Sara	HR	60000	Experience Staff
3	Mark	IT	75000	Staff
4	Lucky	Finance	95000	Staff
5	Tom	HR	55000	Experience HR

If in IT and salary > 80000 Senior IT

If in HR and salary > 55000 Experience HR

Otherwise 'Staff'

EC - emp name
 - Department
 Salary
 - Position level

```

SELECT emp name - position level,
       Salary
CASE
WHEN other wise THEN 'Staff'
WHEN HR & salary > 55000 THEN 'Experience HR'
WHEN IT & salary > 80000 THEN 'Senior IT'
END AS Position level
FROM emp_name
  
```


④ Students

Student id	Student name	Score	Grade
1	Anna	92	A
2	Ben	76	C
3	Cara	59	F
4	David	83	B
5	Ela	68	D

Letter Grade

≥ 90 : 'A'

80-89: 'B'

70-79: 'C'

60-69: 'D'

< 60 : 'F'

EC: Student name

Score

- Grade

```
SELECT Student name,  
Score
```

```
CASE
```

```
WHEN score < 60 THEN 'F'
```

```
WHEN score 60-69 THEN 'D'
```

```
WHEN score 70-79 THEN 'C'
```

```
WHEN score 80-89 THEN 'B'
```

```
WHEN score  $\geq 90$  THEN 'A'
```

```
END AS Grade
```

```
FROM Students;
```


⑤ Deliveries

Delivery id	Delivery time - minutes	Performance
1	45	On time
2	80	Late
3	30	Fast
4	65	Late
5	100	Late

delivery Performance:

≤ 30 mins 'Fast'

31-60 mins 'On time'

60 mins 'Late'

Ec

Delivery id

Delivery time minutes

Performance

```

SELECT Delivery id
        Delivery time minutes
        WHEN min  $\leq$  30 mins 'Late'
        WHEN 31-60 mins 'On time'
        WHEN  $\leq$  30 mins 'Fast'
        END 'Performance'
FROM Deliveries id
  
```


⑥

Tickets

Ticket id	Issue type	Priority	Priority level
1	Login issue	1	Low
2	Server down	3	High
3	Slow system	2	Medium
4	Email error	2	Medium
5	Password reset	1	Low

3 - High

2 - Medium

1 - Low

E.c - Issue type

Priority

Priority label

SELECT issue type,
Priority,

CASE

WHEN ^{Priority level} Priority = 1 THEN 'Low'

WHEN Priority = 2 THEN 'Medium'

WHEN Priority = 3 THEN 'High'

END Priority level

FROM tickets

⑦ Attendance

Student id	days - present	total days	%	Classification
1	45	50	90	Excellent
2	30	50	60	Need imp
3	48	50	96	Excellent
4	25	50	50	Need imp
5	50	50	100	Excellent

$\geq 90\%$ 'Excellent'
 $75-89\%$ 'Good'
 $< 75\%$ 'Needs improvement'

EC Student_d
 Attendance percentage
 Attendance status

```

SELECT Student id
        Attendance Percentage
CASE
WHEN attendance < 75 THEN 'Needs improvement'
WHEN attendance 75-89 THEN 'Good'
WHEN attendance  $\geq 90\%$  THEN 'Excellent'
END Attendance status
FROM Attendance
  
```


⑧ Product inventory

Product id	Stock_qty	Stock status
1	5	In stock
2	0	Out of Stock
3	25	In stock
4	10	In stock
5	3	Low stock

0 'Out of stock'

1-5 'Low stock'

5 'In stock'

E.C Product_id

Stock_qty

Stock status

SELECT Product_id

Stock_qty

CASE

WHEN stock = 0 THEN 'Out of stock'

WHEN stock 1-5 THEN 'Low Stock'

WHEN stock 5 THEN 'In stock'

END Stock status

FROM Product id

⑨ Classes

Class- id	Subject	Enrolled Student	Size
1	Math	30	Large
2	English	25	Large
3	Science	15	Medium
4	Art	5	Small
5	History	20	Medium

$\geq 25 \rightarrow$ 'Large'

$10-24$ Medium

< 10 - Small

EC - Subject

- Enrolled Student

- Class size category

SELECT Subject
Enrolled Student

CASE

WHEN ^{class}size < 10 THEN 'Small'

WHEN class size ^{THEN} $10-24$ 'Medium'

WHEN class size ≥ 25 THEN 'Large'

END Class size category

FROM Classes;

10 Payments

Payment_id	Amount	Payment method	Discount eligibility
1	50	Card	Not eligible
2	200	Cash	Eligible for Discount
3	150	Card	Not eligible
4	75	Paypal	Not eligible
5	300	Cash	Eligible for Discount

EC Payment_id
 Payment method
 Amount
 Discount eligibility

```
SELECT Payment method,  
        Amount,  
        CASE  
            WHEN payment_method = 'cash' >= 200 THEN  
                'Eligible for Discount'  
            WHEN Otherwise 'Not Eligible'  
        END Discount eligibility  
FROM Payments
```